

HOUSE PRICE PREDICTION

Exploratory Data Analysis (EDA) Report

TA-008: Analyze Relationships Between Property Features and House Price

Project Name	House Price Prediction
Assigned Task	TA-008 – Analyze relationships between property features and house price
Dataset	house_price_prediction_dataset.csv
Programming Language	Python
Libraries Used	Pandas and Matplotlib
Prepared By	_____

Submitted for academic/project evaluation

1. Objective

The objective of Task TA-008 is to perform Exploratory Data Analysis (EDA) to understand how important property features are related to house price. The analysis uses numerical and categorical property attributes and visualizes their relationship with the target variable, Price.

2. Dataset Overview

The dataset contains house/property records with the following main attributes:

Feature	Type	Purpose
House_ID	Identifier	Uniquely identifies each house
Square_Feet	Numerical	Represents property size
Bedrooms	Numerical	Number of bedrooms
Bathrooms	Numerical	Number of bathrooms
Floors	Numerical	Number of floors
Year_Built	Numerical	Year in which the house was built
Location	Categorical	Area/location of the property
Condition	Categorical	Overall condition of the house
Price	Numerical / Target	House price to be analyzed

3. Tools and Methodology

Python was used for the analysis. Pandas was used for grouping and calculating average prices, while Matplotlib was used to create charts.

- Grouped property records by Bedrooms, Bathrooms, Floors, Location, and Condition.
- Calculated average house price for the grouped categories.
- Used a scatter plot to study the relationship between Square_Feet and Price.
- Used a scatter plot of average price by Year_Built to observe the time-related pattern.
- Compared the visual trends to identify which property characteristics are associated with differences in house price.

4. EDA Results and Analysis

4.1 Relationship Between Bedrooms and House Price

The bar chart shows that average house price generally increases as the number of bedrooms increases. Houses with 1 bedroom have the lowest average price among the displayed groups, while houses with 4 and 5 bedrooms have the highest average prices. The difference between the 4- and 5-bedroom groups is small.

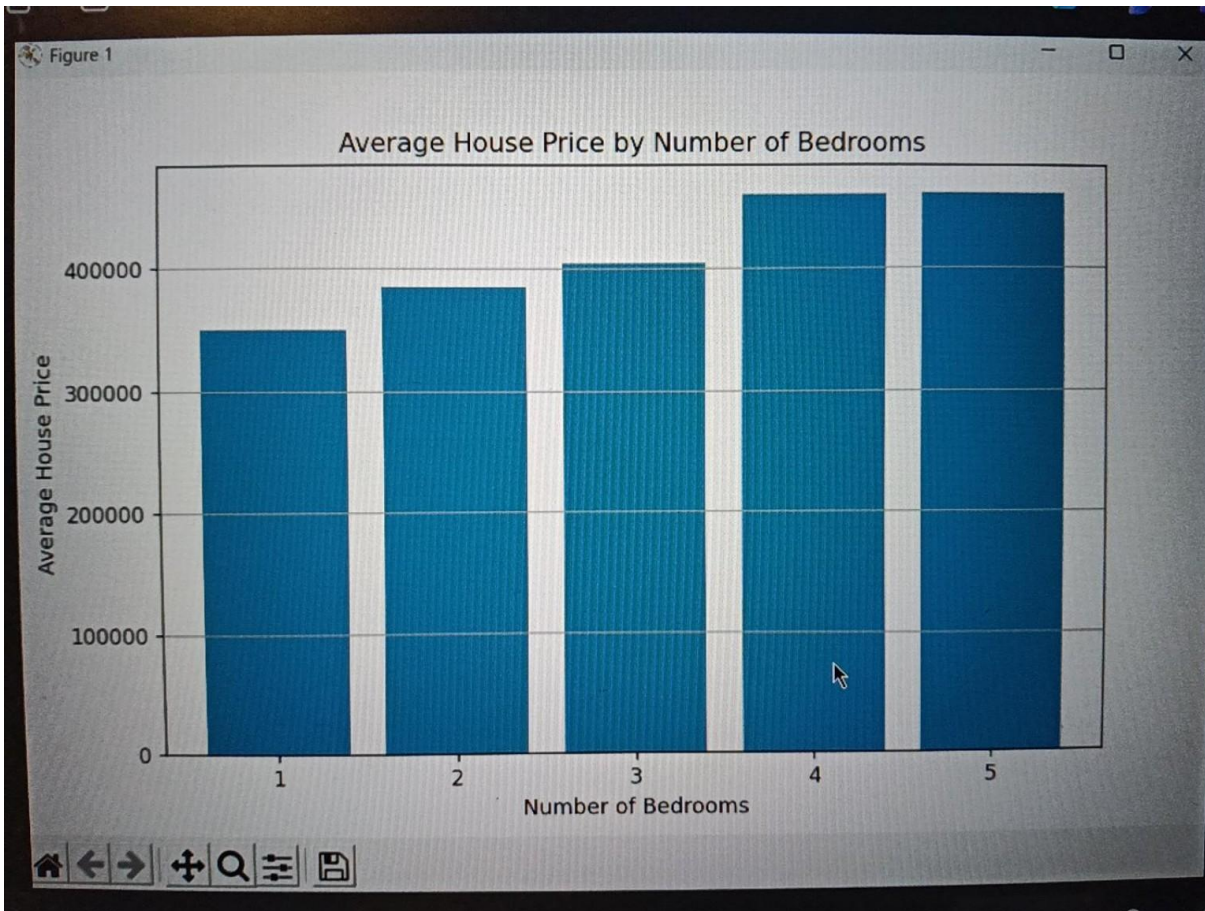


Figure 1. Average House Price by Number of Bedrooms

4.2 Relationship Between Square Feet and House Price

The scatter plot shows a clear positive relationship between property size and price. In general, larger houses tend to have higher prices. The points are spread around the overall upward pattern, which indicates that square footage is an important factor but does not explain house price by itself. Some larger properties have relatively lower or higher prices, suggesting that other features also affect price.

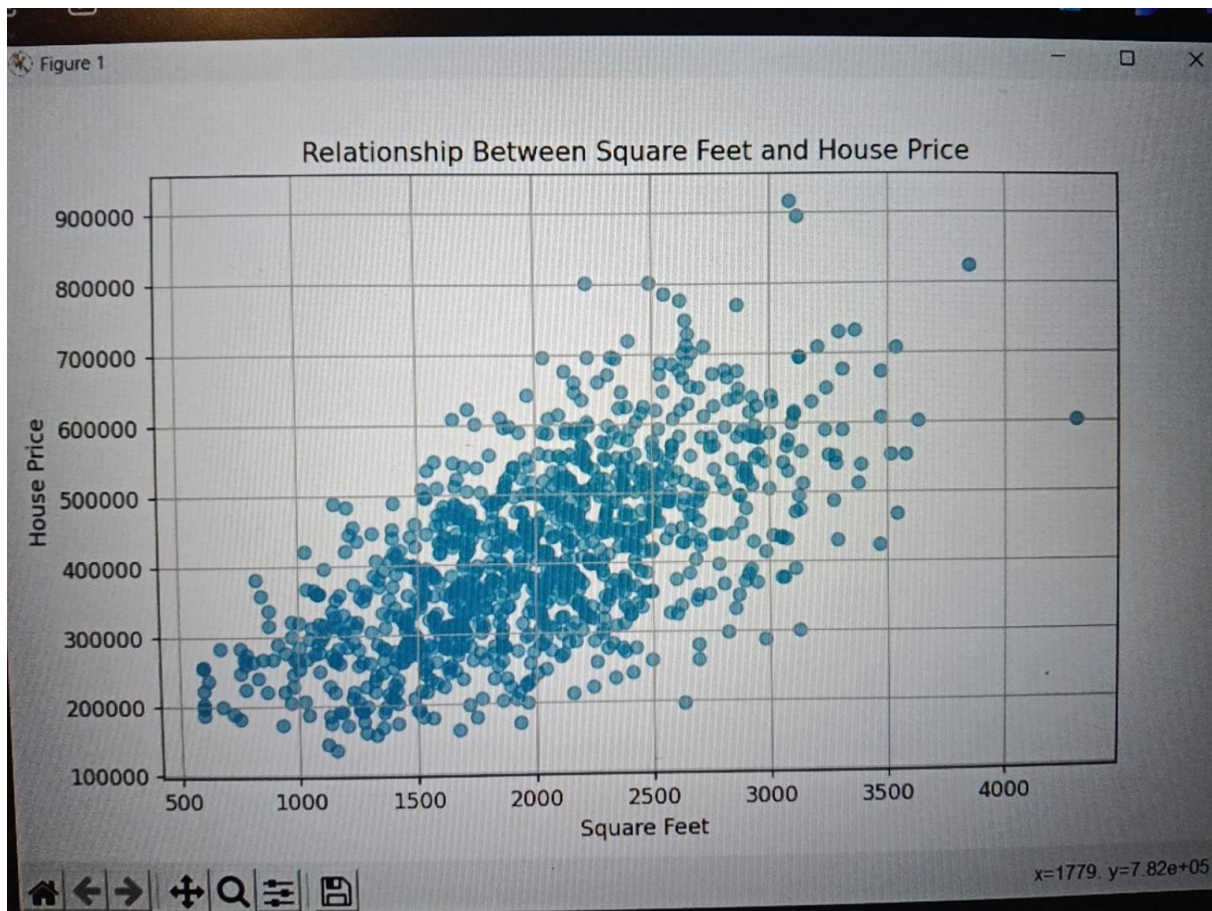


Figure 2. Relationship Between Square Feet and House Price

4.3 Relationship Between Bathrooms and House Price

The chart indicates an increasing pattern in average house price as the number of bathrooms increases. Properties with 1 bathroom have a lower average price, while properties with 4 bathrooms show the highest average price in the chart.

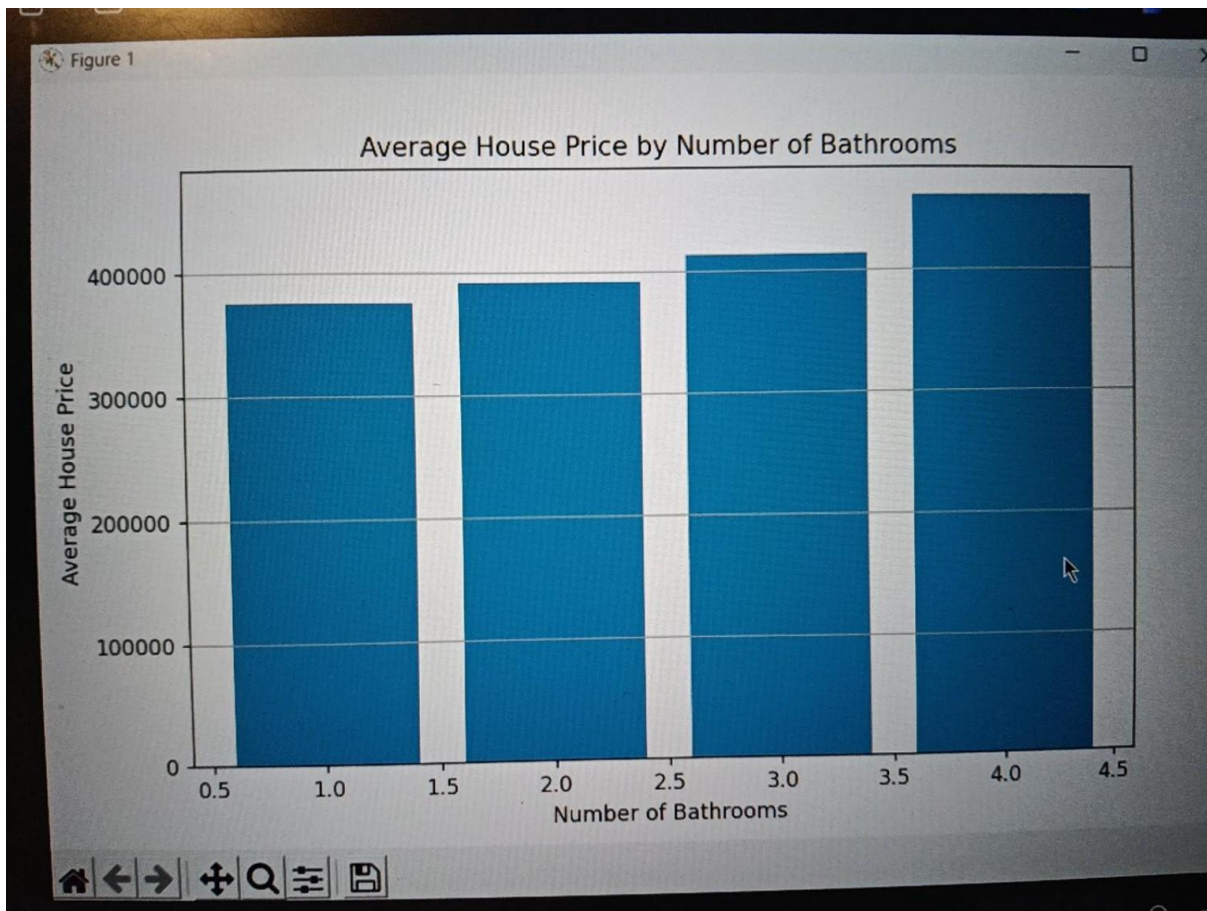


Figure 3. Average House Price by Number of Bathrooms

4.4 Relationship Between Floors and House Price

The average prices for 1, 2, and 3 floors are quite close. The 2-floor category appears slightly higher than the 1-floor and 3-floor categories. Therefore, in this dataset, the number of floors shows a relatively weak difference in average price compared with factors such as size, location, or condition.

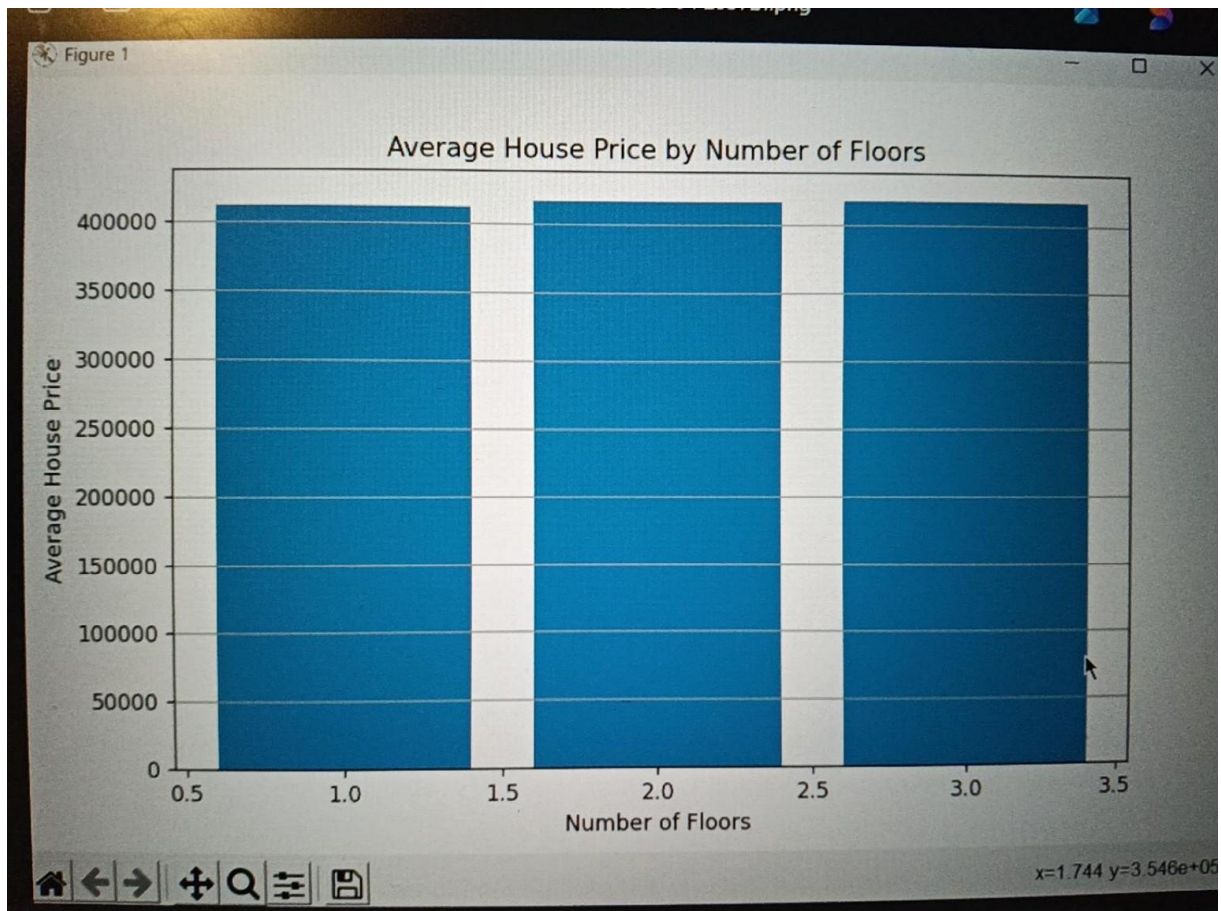


Figure 4. Average House Price by Number of Floors

4.5 Relationship Between Year Built and Average House Price

The scatter plot suggests a mild upward pattern between Year_Built and average house price. Many newer houses are associated with prices around the higher range, although the points show considerable variation. There is also an older-year observation with an unusually high average price, showing that year built alone does not determine price.

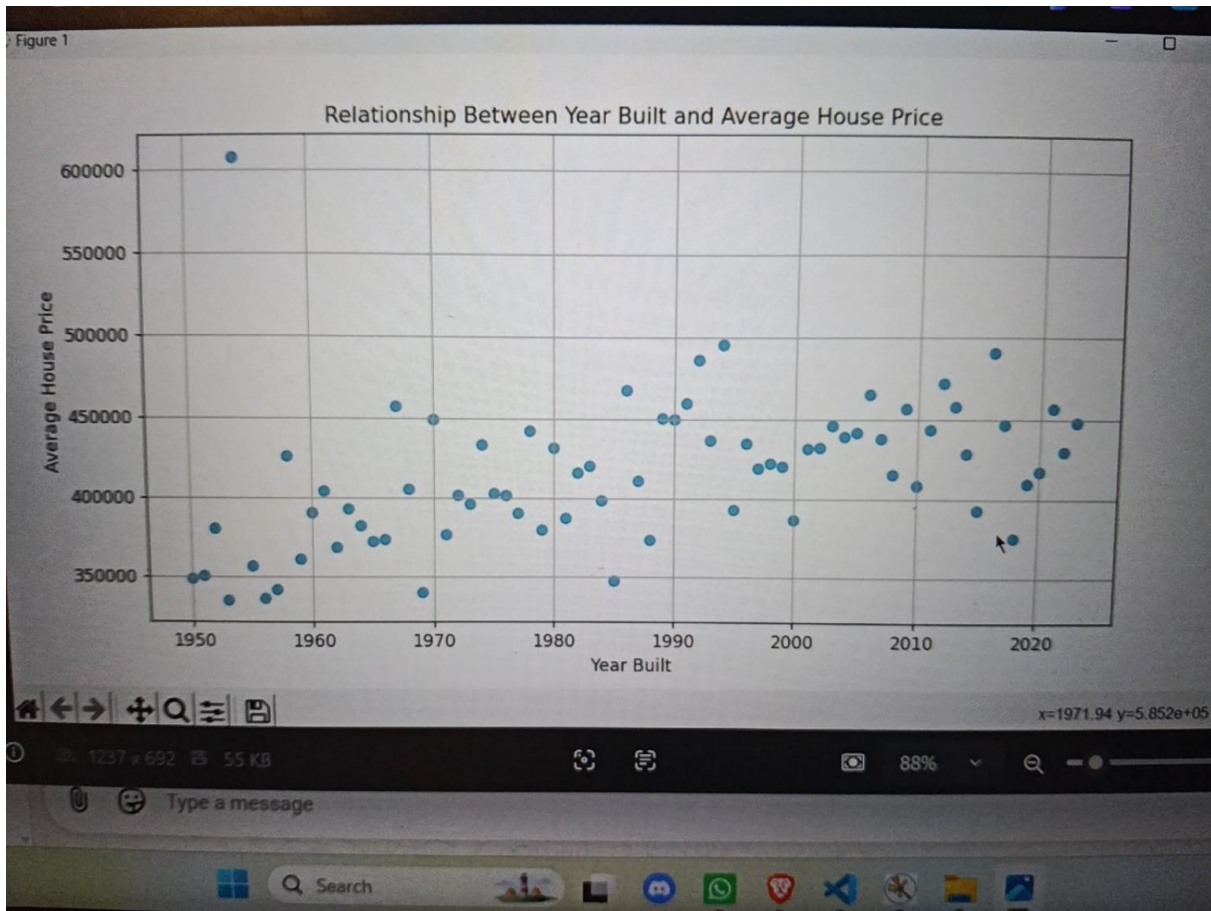


Figure 5. Relationship Between Year Built and Average House Price

4.6 Relationship Between Location and House Price

Location shows a strong difference in average house price. Downtown has the highest average price among the displayed locations. Urban and Suburbs follow, while Rural has the lowest average price. This indicates that location is an important factor associated with house price in this dataset.



Figure 6. Average House Price by Location

4.7 Relationship Between House Condition and House Price

The house-condition chart shows a clear difference in average price between condition categories. Excellent-condition houses have the highest average price, followed by Good, Fair, and Poor. This suggests that better property condition is associated with higher house prices in the dataset.

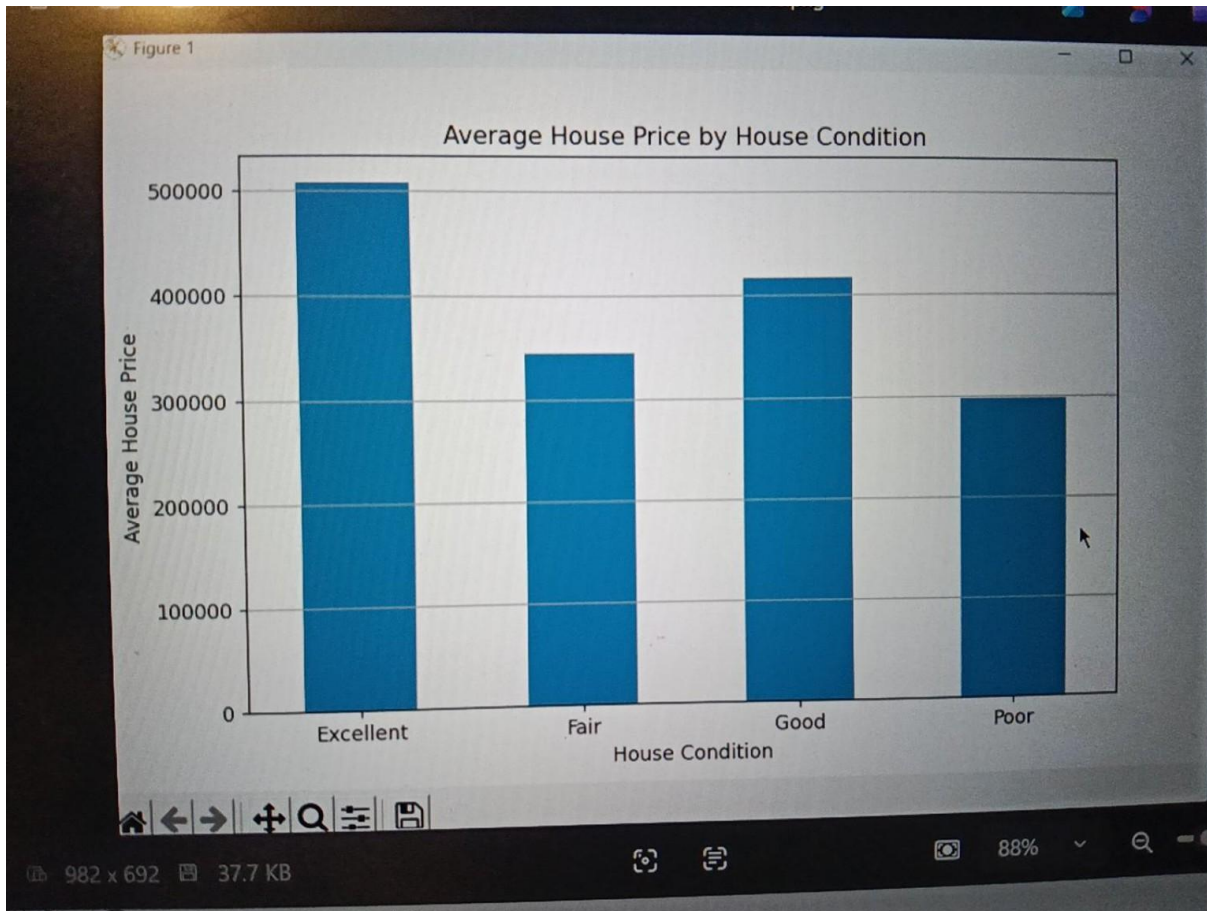


Figure 7. Average House Price by House Condition

5. Summary of Findings

Property Feature	Observed Relationship	Overall Interpretation
Square Feet	Positive relationship	Larger properties generally have higher prices.
Bedrooms	Generally increasing	More bedrooms are associated with higher average prices.
Bathrooms	Increasing	More bathrooms are associated with higher average prices.
Floors	Small variation	Average prices across floor categories are relatively similar.
Year Built	Mild upward pattern	Newer properties tend to be in higher price ranges, with variation.
Location	Strong differences	Downtown has the highest average price; Rural has the lowest.
Condition	Clear increasing pattern	Better condition is associated with higher average price.

6. Key Findings

- Square_Feet shows a clear positive relationship with house price.
- Average price generally increases with the number of bedrooms.
- Average price also increases as the number of bathrooms increases.
- The number of floors produces only small differences in average price in the shown data.
- Year_Built shows a mild upward pattern, but there is noticeable variation.
- Location produces large differences in average house price, with Downtown being the highest among the displayed categories.
- House condition is strongly associated with price: Excellent-condition properties have the highest average price and Poor-condition properties the lowest.

7. Conclusion

The EDA shows that several property features are associated with house price. Square footage, bedrooms, and bathrooms show generally positive relationships with price. Location and house condition show particularly clear differences in average price, while the number of floors has comparatively small differences. Year built shows a mild upward pattern with considerable variation. Overall, the analysis provides useful evidence about which property characteristics may be important for the next stages of the House Price Prediction project.